

Assignment 8: ADF Data Flow 3 Question (Tough)

Advanced Questions :

1. Create a pipeline to copy the customer data from csv file to SQL table which has only 3 columns. While copying the data, ensure that copy all the rows where custid is odd number. Also check if cust id exist in table then update it.

File attached with question.

Table: Create dummy table with column (CID, Name, mailid)

Insert dummy row with id 1,3,5,7,9,8,10

2. Create a pipeline to implement the SCD Type 3 example using the Dataflow.

3. Create a pipeline to remove the duplicate record available in the CSV file and save it as JSON file.a

CSV file attached: SalesLT.Customer.xRyuaEu2p7

.....

Solution:

Question1:

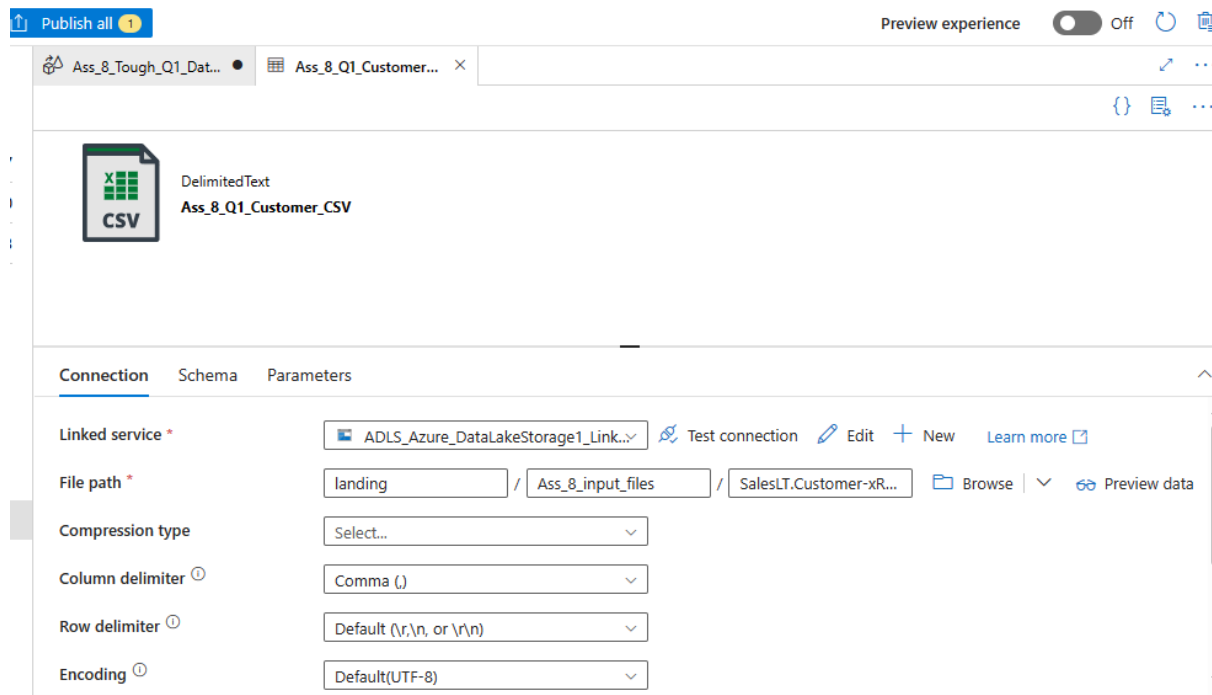
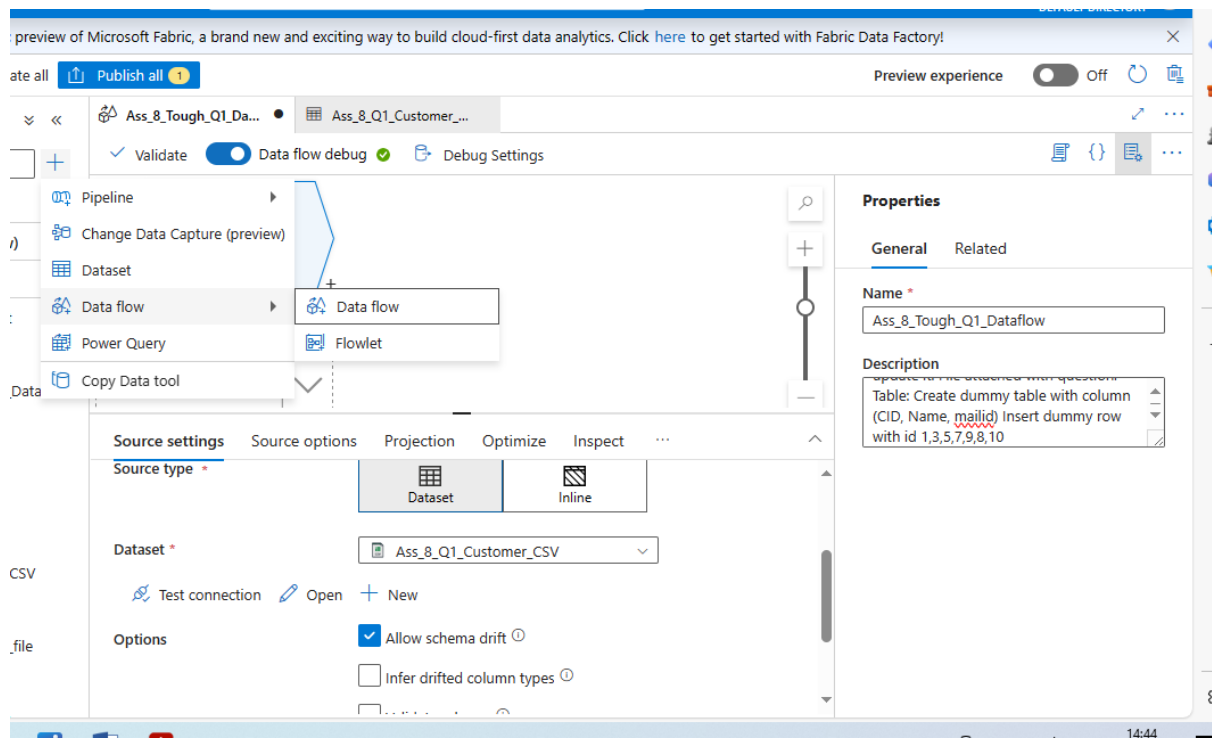
1. Create a pipeline to copy the customer data from csv file to SQL table which has only 3 columns. While copying the data, ensure that copy all the rows where custid is odd number. Also check if cust id exist in table then update it.

File attached with question.

Table: Create dummy table with column (CID, Name, mailid)

Insert dummy row with id 1,3,5,7,9,8,10

Solution:



Customer CSV file in location: (omma separated)

rs > landing >

«

Ass_8_input_files/SalesLT.Customer-xRyuaEu2p7-WdCb... ×

Blob

Save Discard Download Refresh Delete

Overview Versions Edit Generate SAS

1 CustomerID,NameStyle,Title,FirstName,MiddleName,LastName,Suffix,CompanyName,Sales
2 1,False,"Mr.,","Orlando","N.,","Gee",,"A Bike Store","adventure-works\pamela0","orl
3 2,False,"Mr.,","Keith",,"Harris",,"Progressive Sports","adventure-works\david8","k
4 3,False,"Ms.,","Donna",,"F.,","Carreras",,"Advanced Bike Components","adventure-work
5 4,False,"Ms.,","Janet",,"M.,","Gates",,"Modular Cycle Systems","adventure-works\jill
6 5,False,"Mr.,","Lucy",,"Harrington",,"Metropolitan Sports Supply","adventure-works
7 6,False,"Ms.,","Rosmarie",,"J.,","Carroll",,"Aerobic Exercise Company","adventure-wo
8 7,False,"Mr.,","Dominic",,"P.,","Gash",,"Associated Bikes","adventure-works\shu0","d
9 10,False,"Ms.,","Kathleen",,"M.,","Garza",,"Rural Cycle Emporium","adventure-works\j
10 11,False,"Ms.,","Katherine",,"Harding",,"Sharp Bikes","adventure-works\josé1","kat
11 12,False,"Mr.,","Johnny",,"A.,","Caprio",,"Jr.,","Bikes and Motorbikes","adventure-wor
12 16,False,"Mr.,","Christopher",,"R.,","Beck",,"Jr.,","Bulk Discount Store","adventure-w
13 18,False,"Mr.,","David",,"J.,","Liu",,"Catalog Store","adventure-works\michael9","da
14 19,False,"Mr.,","John",,"A.,","Beaver",,"Center Cycle Shop","adventure-works\pamela0
15 20,False,"Ms.,","Jean",,"P.,","Handley",,"Central Discount Store","adventure-works\d
16 21,False,,,"Jinghao",,"Liu",,"Chic Department Stores","adventure-works\jillian0",

Text Preview

Authentication method: Access key (Switch to Microsoft Entra user account)
Location: landing / Ass_8_input_files
Search blobs by prefix (case-...)
Show deleted objects

Name

[-] SalesLT.Customer-xRyuaEu2p7- ***

SalesLT.CustomerAddress-nnC... ***

SalesLT.Product-760RKVH4kL-... ***

source1
Columns: 15 total

filter1
Add expression for the filter

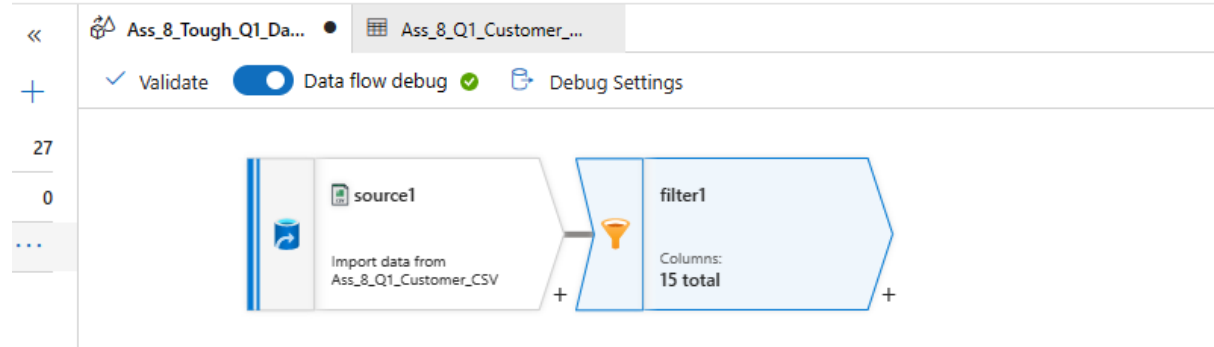
Source settings Source options Projection Optimize Inspect Data preview ●

Number of rows + INSERT 100 * UPDATE 0 × DELETE

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

↑↓	Custom...	12s ↑↓	NameSt...	↑↓	Title	abc ↑↓	FirstName	abc ↑↓	Middle...	abc ↑↓	LastName
+	1	×			Mr.		Orlando		N.		Gee
+	2	×			Mr.		Keith		NULL		Harris
+	3	×			Ms.		Donna		F.		Carreras
+	4	×			Ms.		Janet		M.		Gates

ensure that copy all the rows where custid is odd number.



iset

Filter settings Optimize Inspect Data preview ●

Output stream name * filter1 [Learn more](#)

Description Add expression for the [filter](#) [Reset](#)

Incoming stream * source1

Filter on * Enter filter... ANY

Filter settings Optimize Inspect Data preview ●

Output stream name filter1 [Learn more](#)

Description  copy all the rows where custid is odd number. [Reset](#)

Incoming stream * source1

Filter on * Enter filter... ANY

[Open expression builder](#)

Dataflow expression builder

🔔 filter1

Expression

(CustomerID % 2) != 0

+ - * / || && ! ^ == ===

Expression elements

All

Functions

Input schema

Parameters

Cached lookup

Data flow library functions

Expression values

🔍 Filter by keyword

+ Create new ▾

123 CustomerID

✕ NameStyle

abc Title

Data preview

🔄 Refresh

Save and finish

Cancel

Clear contents

Filter settings Optimize Inspect Data preview ●

Output stream name * [Learn more](#)

Description [Reset](#)

Incoming stream *

Filter on * [x](#)

Ass_8_Tough_Q1_Da... Ass_8_Q1_Customer_...

✓ Validate Data flow debug ● Debug Settings

Filter settings Optimize Inspect **Data preview ●**

Number of rows **INSERT** 100 **UPDATE** 0 **DELETE** 0

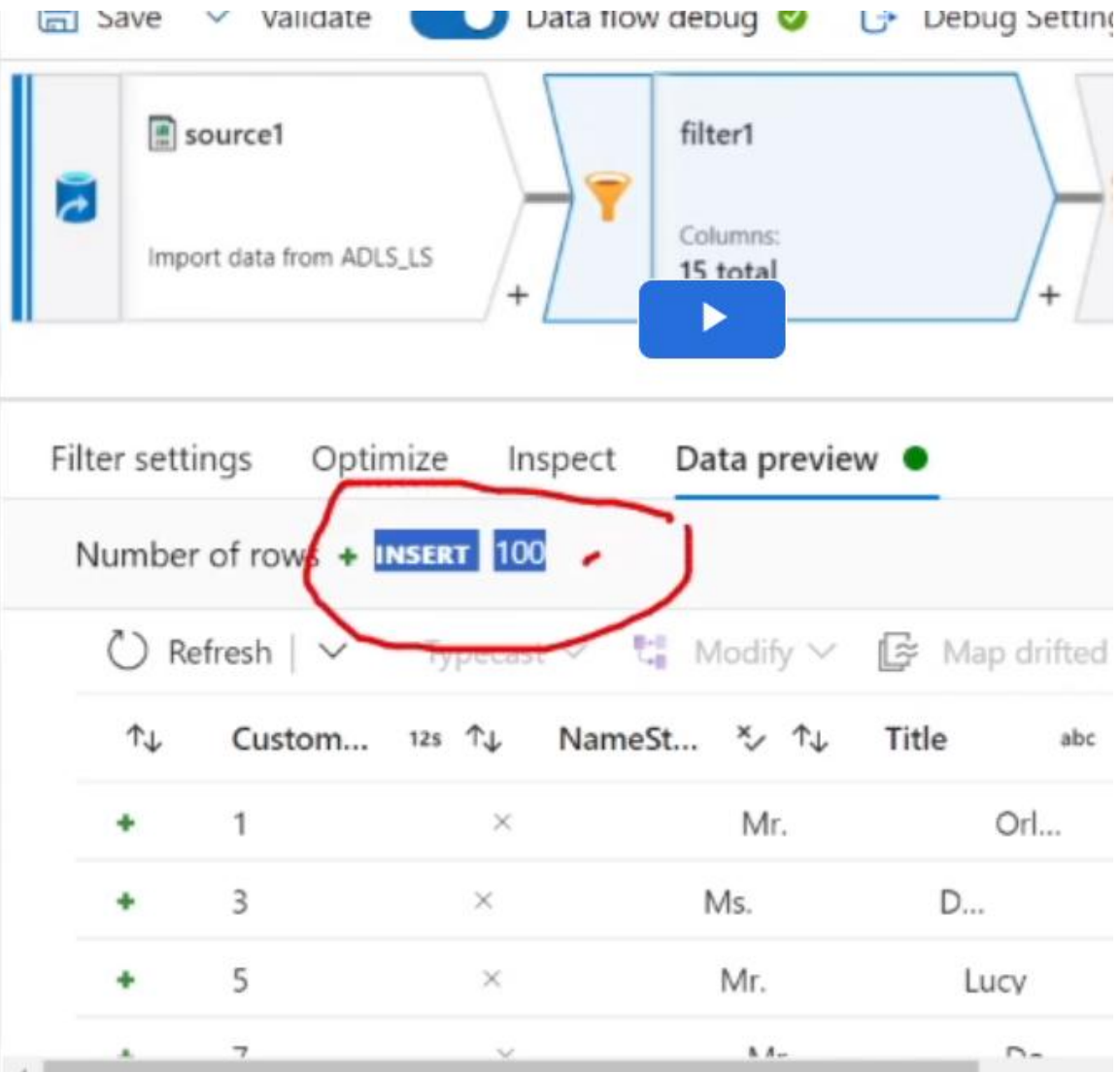
Refresh | Typecast | Modify | Map drifted | Statistics | Remove | Export to CSV

↑↓	Custom...	12s	↑↓	NameSt...	↑↓	Title	abc	↑↓	FirstName	abc	↑↓	Middle...	abc	↑↓	LastName	abc	↑↓
+	1			×		Mr.			Orlando			N.			Gee		
+	3			×		Ms.			Donna			F.			Carreras		
+	5			×		Mr.			Lucy			NULL			Harringt...		
+	7			×		Mr.			Dominic			P.			Gash		

Also check if cust id exist in table then update it.:

Note all the elements in screenshot has + icon which means all are eligible only for **+INSERT**, But not for **"UPSERT"** – SO I USE **"Alter Row"** condition and choose property as **"UPSERT"**

4



Save Validate Data flow debug Debug Settings

source1
Import data from ADLS_LS

filter1
Columns: 15 total

Filter settings Optimize Inspect Data preview

Number of rows + INSERT 100

Refresh Typecast Modify Map drifted

↑↓	Custom...	12s	↑↓	NameSt...	✓	↑↓	Title	abc
+	1		×				Mr.	Orl...
+	3		×				Ms.	D...
+	5		×				Mr.	Lucy
+	7		×				Mr.	D...

pipeline3 • dataflow2 • pipeline4 • Q1

Save ✓ Validate Data flow debug ✓ Debug Settings

source1 filter1 alterRow1

Filtering rows using Columns:

Alter row settings Optimize Insp [Data preview](#) ●

Output stream name * alterRow1 [Learn more](#) [Reset](#)

Description Add expressions to alter rows

Incoming stream * filter1

Alter row conditions * ① Upsert if Enter condition...

Give Upsert IFF "1==1", so all row eligible for upsert

4

source1 filter1 alterRow1

Filtering rows using Columns:

Alter row settings Optimize Insp [Data preview](#) ●

Output stream name * alterRow1 [Learn more](#) [Reset](#)

Description Add expressions to alter rows

Incoming stream * filter1

Alter row conditions * ① Upsert if 1==1 [Open expression builder](#)

view of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

ill Publish all

Preview experience Off

Ass_8_Tough_Q1_Da... • Ass_8_Q1_Customer_...

✓ Validate Data flow debug Debug Settings

27

0

23

Alter row settings Optimize Inspect **Data preview**

Number of rows INSERT 0 UPDATE 0 DELETE 0

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

	Custom...	12s	NameSt...	Title	abc	FirstNa...	abc	Middle...	abc	LastName	abc
	1		Mr.	Orlando	N.	Gee					
	3		Ms.	Donna	F.	Carreras					
	5		Mr.	Lucy	NULL	Harringt...					
	7		Mr.	Dominic	P.	Gash					

Upsert

pipeline3 • dataflow2 • pipeline4 • Q1 •

Save ✓ Validate Data flow debug Debug Settings

Alter row settings Optimize Inspect **Data preview**

Output stream name * alterRow1 [Learn more](#)

Description Reset

Incoming stream * filter1

Alter row conditions * Upsert if 1==1

csv file to SQL table which has only 3 columns.



Select settings Optimize Inspect Data preview ●

Output stream name *

select1

[Learn more](#)

Description

Renaming alterRow1 to select1 with columns 'CID, Name, mailid'

[Reset](#)

Incoming stream *

alterRow1

Options

☒ Skip duplicate input columns ⓘ

☒ Skip duplicate output columns ⓘ

↑ Publish all 3 Preview e

Ass_8_Tough_Q1_Da... Ass_8_Q1_Customer... SQL_Address_table_DS Cust_dummy_SQLtab...

✓ Validate Data flow debug Debug Settings

Add Source

Select settings Optimize Inspect **Data preview**

Number of rows INSERT 0 UPDATE 0 DELETE 0 UPSERT 100 LOOKUP 0

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

↑↓	CID	12s ↑↓	Name	abc ↑↓	mailId	abc ↑↓
✱	1		Orlando		orlando0@adventur...	
✱	3		Donna		donna0@adventure...	
✱	5		Lucy		lucy0@adventure-w	

File attached with question.

Table: Create dummy table with column (CID, Name, mailid)

Insert dummy row with id 1,3,5,7,9,8,10

Query to create and insert Dummy Table:

```
Create table [SalesLT].[Ass_8_Tough_Q1_Cust_Dummy_table]
(CID integer, Name varchar(100), mailid varchar(100))
```

```
Insert into [SalesLT].[Ass_8_Tough_Q1_Cust_Dummy_table] (CID)
Values
(1),(3),(5),(7),(9),(8),(10);
```

```
Select * from [SalesLT].[Ass_8_Tough_Q1_Cust_Dummy_table]
```

```
Select * from [SalesLT].[Cust2Table_Ass8_part2_Q1]
```


012024db (mission100gopi01012024dbserver1/mission100Gopi01012024db)

mission100Gopi01012024db (mission100gopi01012024dbserver1/mission1... SQL database

Search Login New Query Open query Feedback Getting started

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)

Settings
Compute + storage
Connection strings
Properties
Locks
Data management
Replicas
Sync to other databases

Query 1 X

Run Cancel query Save query Export data as Show only Editor

```

2 Create table [SalesLT].[Ass_8_Tough_Q1_Cust_Dummy_table]
3 (CID integer, Name varchar(100), mailid varchar(100))
4
5 Insert into [SalesLT].[Ass_8_Tough_Q1_Cust_Dummy_table] (CID)
6 Values
7 (1),(3),(5), (7),(9),(8),(10);
8
9 Select * from [SalesLT].[Ass_8_Tough_Q1_Cust_Dummy_table]

```

Results Messages

1
3
5
7
9

mission100Gopi01012024db (mission100gopi01012024dbserver1/mission1... SQL database

Search Login New Query Open query Feedback Getting started

Overview
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Query editor (preview)

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Sync to other databases
Integrations

mission100Gopi01012024db (az... Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

Tables

- > dbo.BuildVersion
- > dbo.ErrorLog
- > dbo.HW_table
- > SalesLT.Address
- > SalesLT.Ass_8_Tough_Q1_Cust_D
- > SalesLT.Customer
- > SalesLT.CustomerAddress
- > SalesLT.CustomerClone
- > SalesLT.Product
- > SalesLT.ProductCategory
- > SalesLT.ProductDescription

Query 1 X

Run Cancel query Save query

```

2 Create table [SalesLT].[Ass_8_Tough_Q1_Cust
3 (CID integer, Name varchar(100), mailid var
4
5 Insert into [SalesLT].[Ass_8_Tough_Q1_Cust
6 Values
7 (1),(3),(5), (7),(9),(8),(10);
8
9 Select * from [SalesLT].[Ass_8_Tough_Q1_Cus

```

Result Messages

1
3
5
7
9

Query succeeded | 0s

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

gopinathma09@outlook

DEFAULT DIRECTORY (ANIL24SH)

> SQL databases > mission100Gopi01012024db (mission100gopi01012024dbserver1/mission100Gopi01012024db)

L databases

t Directory (Anil24sharmaoutlook.onmicros...

Create Reservations

for any field...

↑↓

mission100Anil06121992DB (mission10...

mission100Gopi01012024db (mission10...

mission100ila01012024db (mission100i...

mission100Gopi01012024db (mission100gopi01012024dbserver1/mission1...

SQL database

Search

Login New Query Open query Feedback Getting started

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Settings

Compute + storage

Connection strings

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Data management

Replicas

Sync to other databases

Integrations

Query 2

Run Cancel query Save query Export data as Show only Editor

1 Select * from [SalesLT].[Ass_8_Tough_Q1_Cust_Dummy_table]

2

3 Select * from [SalesLT].[Cust2Table_Ass8_part2_Q1]

Results

Messages

Failed to execute query. Error: Invalid object name 'SalesLT.Cust2Table_Ass8_part2_Q1'.

Query failed

source1
Import data from Ass8_Q1_Customer.CSV

filter1
Filtering rows using expression based on column 'CustomerID'

sink1
Columns: 15 total

Add Source

Now Sink Type is DB

Sink Settings Errors Mapping Optimize Inspect Data preview

Output stream name * sink1 [Learn more](#)

Description Add sink dataset [Reset](#)

Incoming stream * filter1

Sink type * Dataset Inline Cache

Dataset * Select... [+ New](#)

Options ☒ Allow schema drift [?](#) ☐ Validate schema [?](#)

Create a pipeline to copy the customer data from csv file to **SQL** where the customer id is an even number. (Attaching the Customer File)

File: SalesLT.Customer-xRyuaEu2p7-WdCbzuWllo-80V7BfCQtP

DEVELOPER

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of data within a data store. [Learn more](#)

Select a data store

sql

All

Azure

Database

File

Generic protocol

NoSQL

Services and applications



Azure Cosmos DB for NoSQL



Azure Database for MySQL



Azure Database for PostgreSQL



Azure SQL Database



Azure SQL Database Managed Instance



SQL server



Amazon RDS for SQL Server



MySQL



PostgreSQL

Continue

gopinathma09@outlook.co
DEFAULT DIRECTO

Set properties

Name
SQLDB_CustomerDB

Linked service *
SQLDB_Linked_service1

☐ Select from existing table ☒ New table

Schema and table name
SalesLT . CustomerClone

> Advanced

Publish all 4

Ass_8_Tough_Q1_Dat... Ass_8_part2_Q1_To... AzureSqlTable1

 Azure SQL Database
AzureSqlTable1

Connection Schema Parameters

Linked service * SQLDB_Linked_service1

Test connection Edit + New Learn more

Table SalesLT.Cust2Table_Ass8_part2_Q1 Refresh

☐ Enter manually

Preview data

Microsoft Azure Upgrade Search resources, services, and docs (G+/)

SQL databases > mission100Gopi01012024db (mission100gopi01012024dbserver1/mission100Gopi01012024db)

L databases

t Directory (Anil24sharmaoutlook.onmicros...

Create Reservations ...

for any field...

↑ ↓

mission100Anil06121992DB (mission10...
mission100Gopi01012024db (mission10...
mission100ila01012024db (mission100i...

mission100Gopi01012024db (mission100gopi01012024dbserver1/mission1... ☆ ...

SQL database

Search Login + New Query ↑ Open query Feedback Getting started

- Overview
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- Query editor (preview)

Settings

- Compute + storage
- Connection strings
- Properties
- Locks

Data management

- Replicas
- Sync to other databases

Integrations

Page 1 of 1

Query 2

Run Cancel query Save query Export data as Show only Editor

```
1 Select * from [SalesLT].[Ass_8_Tough_Q1_Cust_Dummy_table]
2
3 Select * from [SalesLT].[Cust2Table_Ass8_part2_Q1]
```

Results Messages

Failed to execute query. Error: Invalid object name 'SalesLT.Cust2Table_Ass8_part2_Q1'.

Query failed

Validate Data flow debug Debug Settings

source1 Import data from Ass_8_Q1_Customer_CSV

filter1 copy all the rows where custid is odd number.

alterRow1 Add expressions to alter rows

select1 Renaming alterRow1 select1 with columns Name, mailid

Add Source

Sink Settings Errors Mapping Optimize Inspect Data preview

Output stream name * sink1 Learn more

Description Export data to AzureSqlTable1 Reset

Incoming stream * select1

Sink type * Dataset Inline Cache

ew of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

[Publish all](#) 4 Preview experience Off 🔄 🗑️

Ass_8_Tough_Q1_Da... • Ass_8_part2_Q1_To...

✓ Validate 🔧 Data flow debug 🟢 Debug Settings 📄 { } 🔍 ⋮

Reference: 1
Columns: 15 total

filter1
copy all the rows where custid is odd number.

alterRow1
Add expressions to alter rows

select1
Renaming alterRow1 to select1 with columns 'CID, Name, mailid'

sink1
Columns: 3 total

Sink Settings Errors Mapping Optimize Inspect Data preview ●

Incoming stream * select1

Sink type * Dataset Inline Cache

Dataset * AzureSqlTable1 🔗 Test connection 🔗 Open ➕ New

Options ☒ Allow schema drift ⓘ ☐ Validate schema ⓘ

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Ass_8_Tough_Q1_Da... • Ass_8_part2_Q1_To...

✓ Validate 🔧 Data flow debug 🟢 Debug Settings 📄 { } 🔍 ⋮

source1
Import data from Ass_8_Q1_Customer_CSV

filter1
copy all the rows where custid is odd number.

alterRow1
Add expressions to alter rows

select1
Renaming alterRow1 to select1 with columns 'CID, Name, mailid'

sink1
Columns: 3 total

Sink Settings Errors Mapping Optimize Inspect Data preview ●

Output stream name * sink1 [Learn more](#) 🔗

Description Export data to AzureSqlTable1 🔄 Reset

Incoming stream * select1

Sink type * Dataset Inline Cache

Dataset * AzureSqlTable1 🔗 Test connection 🔗 Open ➕ New

17:57

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Preview experience Off

Ass_8_Tough_Q1_Da...

Ass_8_part2_Q1_To...

Validate

Data flow debug

Debug Settings

Reference: 1
Columns: 15 total

filter1
copy all the rows where custid is odd number.

alterRow1
Add expressions to alter rows

select1
Renaming alterRow1 to select1 with columns 'CID, Name, mailid'

Columns: 3 total

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Update method

Allow insert

Allow delete

Allow upsert

Allow update

Key columns

List of columns

Custom expression

123 CID

Add Source

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Schema

Input

Output

Number of columns

Total

3

Order	Column	Type
1	CID	12s short
2	Name	abc string
3	mailid	abc string

Ass_8_Tough_Q1_Da... Ass_8_part2_Q1_To...

Validate Data flow debug Debug Settings

source1 Import data from Ass_8_Q1_Customer_CSV

filter1 copy all the rows where custid is odd number.

alterRow1 Add expressions to alter rows

select1 Renaming alterRow1 to select1 with columns 'CID, Name, mailid'

sink1 Columns: 3 total

Sink Settings Errors Mapping Optimize Inspect Data preview

Schema Input Output

Number of columns	Updated 0	Dropped 0	Unchanged 3	Total 3
Order	Column	Type	Updated	Input column
1	CID	12s short		CID
2	Name	abc string		Name
3	mailid	abc string		mailid

17:52

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Publish all 4 Preview experience Off

Ass_8_Tough_Q1_Da... Ass_8_part2_Q1_To...

Validate Data flow debug Debug Settings

source1 Import data from Ass_8_Q1_Customer_CSV

filter1 copy all the rows where custid is odd number.

alterRow1 Add expressions to alter rows

select1 Renaming alterRow1 to select1 with columns 'CID, Name, mailid'

sink1 Columns: 3 total

Sink Settings Errors Mapping Optimize Inspect Data preview

Number of rows INSERT 0 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL 407

Refresh Statistics Export to CSV

	CID	Name	mailid
1	Orlando	orlando0@adventur...	
3	Donna	donna0@adventure...	
5	Lucy	lucy0@adventure-w...	
7	Dominic	dominic0@adventur...	
11	Katherine	katherine0@advent...	

17:52

Finally Run pipeline and publish all

and clicking may be some steps first data entry then click [here](#) to get started with Azure Data Factory.

Preview experience ☐ Off  

Ass_8_part2_Q1_To...  

«  Validate  Debug   Add trigger ☒ Data flow debug 

 Data flow 

Properties View the JSON code representation of this resource

General Related

Name *
Ass_8_part2_Q1_Tough_pipeline

Description
Create a pipeline to copy the customer data from csv file to SQL table which has only 3 columns. While copying the data,

Annotations
 New

General Settings Parameters User properties

Name *
Data flow
[Learn more](#) 

Description
Create a pipeline to copy the customer data from csv file to SQL table which has only 3 columns. While copying the

Activity state 
☒ Activated ☐ Deactivated

Timeout 

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Preview experience

Ass_8_part2_Q1_To...

✓ Validate

▶ Debug

⚡ Add trigger

☒ Data flow debug

Data flow

 Data flow

General

Settings

Parameters

User properties

Data flow *

Ass_8_Tough_Q1_Dataflow

 Open



Run on (Azure IR) * ⓘ

AutoResolveIntegrationRuntime

Compute size * ⓘ

Small

> Advanced

Logging level * ⓘ

☒ Verbose ☐ Basic ☐ None

> Sink properties

> Staging ⓘ

Preview experience

Ass_8_part2_Q1_To...

⌵ ⏪

✓ Validate

▶ Debug

⚡ Add trigger

☒ Data flow debug

Trigger test runs of the current pipeline without publishing your changes to the service

Data flow

 Data flow

Preview experience

Off

Ass_8_part2_Q1_To...

Validate

Debug

Add trigger

Data flow debug

Data flow

Data flow

Parameters

Variables

Settings

Output

Pipeline run ID: 2a0ac647-23d1-45c2-9254-aba499c6c963

Pipeline status: Succeeded

View debug run consumption

All status

Monitor in Azure Metrics

Export to CSV

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration
Data flow	Succeeded	Data flow	3/3/2024, 5:48:42 PM	4m 18s

Once pipeline success , checkif Table got created and rows got inserted

mission100Gopi01012024db (mission100gopi01012024dbserver1/mission100gopi01012024db) ☆ ... ✕

« Login + New Query ↑ Open query Feedback Getting started

Query 2 ✕

Run ☐ Cancel query Save query Export data as Show only Editor

```
1
2
3 select * from [SalesLT].[Cust2Table_Ass8_part2_Q1]
```

Results Messages

Search to filter items...

CID	Name	mailId
1	Orlando	orlando0@adventure-works.com
3	Donna	donna0@adventure-works.com
5	Lucy	lucy0@adventure-works.com

Query succeeded | 2s

home > SQL databases > mission100Gopi01012024db (mission100gopi01012024dbserver1/mission100Gopi01012024db) | Que... ☆ ... ✕

SQL database

Search

« Login + New Query ↑ Open query Feedback Getting started

mission100Gopi01012024db (azurelib)

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

Tables

- dbo.BuildVersion
- dbo.ErrorLog
- dbo.HW_table
- SalesLT.Address
- SalesLT.Ass_8_Tough_Q1_Cust_Dummy_table**
- SalesLT.Customer
- SalesLT.CustomerAddress
- SalesLT.CustomerClone
- SalesLT.Product
- SalesLT.ProductCategory
- SalesLT.ProductDescription

Query 2 ✕

Run ☐ Cancel query Save query Export data as ...

```
1
2
3 select * from [SalesLT].[Cust2Table_Ass8_part2_Q1]
```

Results Messages

Search to filter items...

CID	Name	mailId
1	Orlando	orlando0@adventur
3	Donna	donna0@adventure
5	Lucy	lucy0@adventure-w

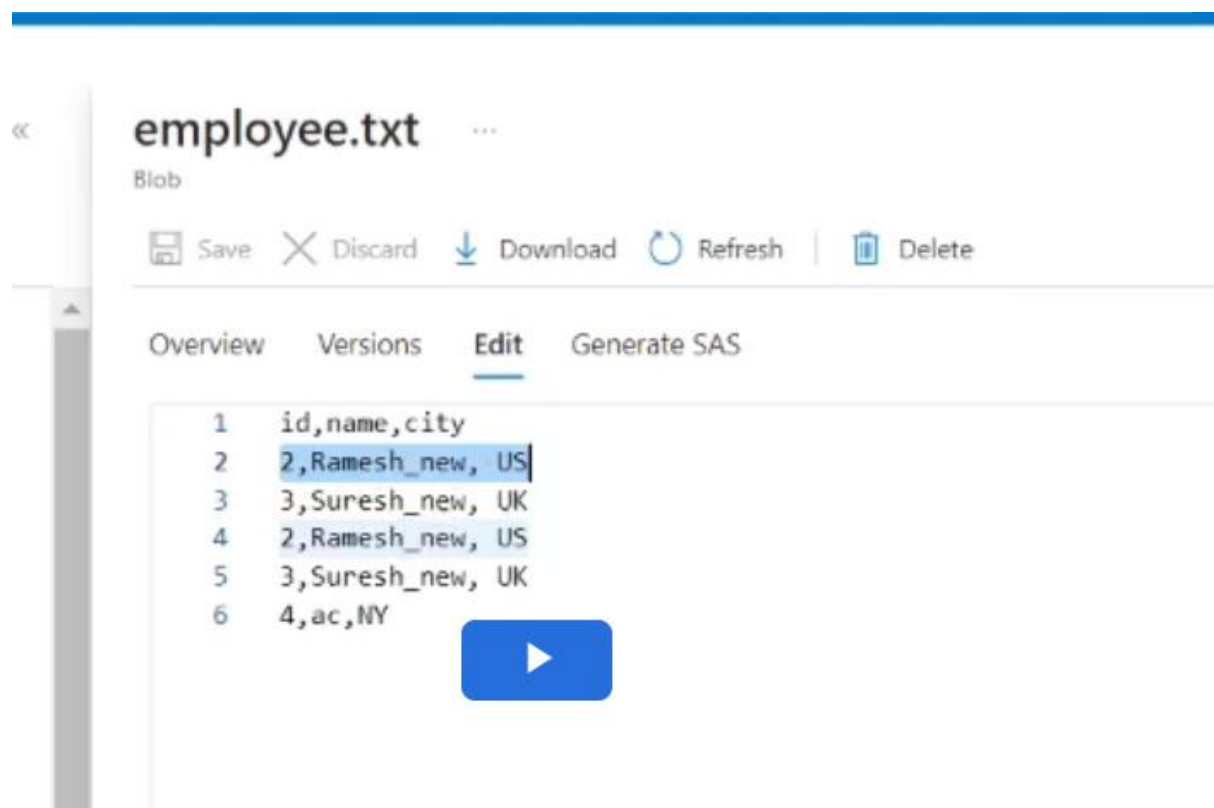
Finally Click Publish all in Pipeline

Question 3:

3. Create a pipeline to remove the duplicate record available in the CSV file and save it as JSON file.a

Solution 3:

First load file into Dataflow and then remove duplicate



Containers > landing >

«

«

Upload

+ Add Directory

...

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing / Ass_8_input_files

Search blobs by prefix (case-...)

Show deleted objects

Name

employee.txt

SalesLT.Customer-xRyuaEu2p7...

SalesLT.CustomerAddress-nnC...

SalesLT.Product-760RKVH4kL-...

Ass_8_input_files/employee.txt

Blob

Save Discard Download Refresh Delete

Overview Versions Edit Generate SAS

1 id,name,city

2 2,Ramesh_new,US

3 3,Suresh_new,UK

4 2,Ramesh_new,US

5 3,Suresh_new,UK

6 4,ac,NY

Text Preview

Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Publish all 1

Preview experience Off

Ass_8_Tough_Q1_Dat...

Ass_8_Part2_Q3_Dat...

Validate

Data flow debug

Debug Settings

Add Source

+

—

□

⌕

Properties

General Related

Name *

Ass_8_Part2_Q3_DataFlow

Description

3. Create a pipeline to remove the duplicate record available in the CSV file and save it as JSON file.a

view of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

all [Publish all](#) 2 Preview experience

« [Ass_8_Tough_Q1_Dat...](#) [Ass_8_Part2_Q3_Dat...](#) • [ADLS_Generic_CSV_DS](#)

✓ Validate ☒ Data flow debug ✓ [Debug Settings](#)



26 [Source settings](#) [Source options](#) [Projection](#) [Optimize](#) [Inspect](#) [Data preview](#) ●

Output stream name * [Learn more](#) 🔗

Description [Reset](#) 🔄

Source type * ☒ Dataset ☐ Inline

Dataset * 📄 ✓ Connection successful [Test connection](#) 🔗 [Open](#) + [New](#)

Options ☒ Allow schema drift 🔗

[Source settings](#) [Source options](#) [Projection](#) [Optimize](#) [Inspect](#) [Data preview](#) ●

[📄 Define default format](#) [🔍 Detect data type](#) [↩ Import projection](#) [🔄 Reset schema](#)

Column name	↑↓ 🔍 Type	↑↓ 🔍 Format
id	abc string	Specify format
name	abc string	Specify format
city	abc string	Specify format

Source settings	Source options	Projection	Optimize	Inspect	Data preview
Number of columns Total 3					
Order	↑↓	Column	↑↓	Type	↑↓
1		id		abc	string
2		name		abc	string
3		city		abc	string

✓ Validate
☒ Data flow debug
🔍 Debug Settings

Source settings
Source options
Projection
Optimize
Inspect
Data preview

Number of rows **INSERT** 5 **UPDATE** 0 **DELETE** 0 **UPSERT** 0

Refresh | Typecast | Modify | Map drifted | Statistics | Remove

↑↓	id	abc	↑↓	name	abc	↑↓	city	abc
+	2			Ramesh new			US	
+	3			Suresh new			UK	
+	2			Ramesh new			US	
+	3			Suresh new			UK	
+	4			ac			NY	

Aggregate on each column will bring duplicate to single unique value

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ste all [Publish all](#) 2 Preview experience ☐ Off

Ass_8_Tough_Q1_Dat... Ass_8_Part2_Q3_Dat... ADLS_Generic_CSV_DS

✓ Validate ☒ Data flow debug ☒ Debug Settings

source1
Import data from Duplicate_Employee_CSV_file

aggregate1
Columns: 4 total

Aggregate settings Optimize Inspect Data preview ●

Description
Aggregating data by 'id, name, city' producing columns 'Max_ID' [Reset](#)

Incoming stream * source1

Group by Aggregates

Columns	Name as		
abc id	id	+	🗑️
abc name	name	+	🗑️
abc city	city	+	🗑️

Fpor namesake, it mis must to use Aggregate function (because we used Group by , so Aggregate is must as per syntax)

So we apply some Maxid or so

validate Data flow debug Debug Settings



Aggregate settings Optimize Inspect Data preview ●

Incoming stream * source1

Group by Aggregates

Grouped by: id, name, city

+ Add Clone Delete Open expression builder

Column	Expression
☐ Max_ID	max(id) abc +

If you see “Data Preview” tab, u get all unique output along with extra column Max ID, so I again use Select function and just use 3 columns as output

validate Data flow debug Debug Settings



Aggregate settings Optimize Inspect Data preview ●

Number of rows + INSERT 3 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

id	name	city	Max_ID
2	Ramesh new	US	2
3	Suresh new	UK	3
4	ac	NY	4

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Publish all (2) Preview experience Off

Ass_8_Tough_Q1_Dat... Ass_8_Part2_Q3_Dat... ADLS_Generic_CSV_DS

Validate Data flow debug Debug Settings

source1 Import data from Duplicate_Employee_CSV_file

aggregate1 Aggregating data by 'id, name, city' producing columns 'Max_ID'

select1 Columns: 3 total

Select settings Optimize Inspect Data preview

Options

- ☒ Skip duplicate input columns
- ☒ Skip duplicate output columns

Input columns *

☐ Auto mapping Reset Add mapping Delete 3 mappings: 1 column(s) from the inputs left unmapped

	aggregate1's column	Name as
☐	abc id	id
☐	abc name	name
☐	abc city	city

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Publish all (2) Preview experience Off

Ass_8_Tough_Q1_Dat... Ass_8_Part2_Q3_Dat... ADLS_Generic_CSV_DS

Validate Data flow debug Debug Settings

source1 Import data from Duplicate_Employee_CSV_file

aggregate1 Aggregating data by 'id, name, city' producing columns 'Max_ID'

select1 Columns: 3 total

Select settings Optimize Inspect Data preview

Number of rows INSERT 3 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

	id	name	city
2		Ramesh new	US
3		Suresh new	UK
4		ac	NY

and save it as JSON file.a → We need Destination as Sink

view of Microsoft Fabric, a brand new and exciting way to build clo

all ↑ Publish all 2

Ass_8_Tough_Q1_Dat... Ass_8_Part2_Q3_Dat...

✓ Validate ☒ Data flow debug ✓ Debu

source1

Import data from Duplicate_Employee_CSV_fil e

aggregate

Aggregating name, city' p columns 'Ma

Sink Settings Errors Mapping Optimiz

Description

Add sink data

Incoming stream *

select1

Sink type *

Dataset

Dataset *

Select...

Options

☒ Allow schema

☐ Validate sche

Select format

Choose the format type of your data

Avro

DelimitedText

JSON

ORC

Parquet

Binary

Continue

Back

Cancel

✓ Validate ☒ Data flow debug ☒ Debug Settings

Sink Settings Errors Mapping Optimize Inspect **Data preview**

Number of rows **INSERT** N/A **UPDATE** N/A **DELETE** N/A **UPSERT** N/A **LOOKUP** N/A **ERROR** N/A **TOTAL**

Refresh | Statistics | Export to CSV

id	name	city
2	Ramesh new	US
3	Suresh new	UK
4	ac	NY

Finally create a Pipeline to Execute a Daaflow:

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Data Factory Validate all Publish all Preview experience Off

Factory Resources

Filter resources by name

Pipelines

- Ass_5_3_JOIN_Ingestion_Custom...
- Ass_6_3_Foreach_Example_2
- Ass_6_4_Foreach_Example_2
- Ass_6_5_Cust_YYYY_MM_DD
- Ass_7_1_Get_MetaData_Example
- Ass_7_4_Table_HighWaterMarkFile
- Ass_7_ques_3
- Ass_7_ques_5_Foreach_Example2
- Ass_8_part2_Q1_Tough_pipeline
- Ass_8_Q2_Pipeline_Dataflow
- Ass_8_Q3_Pipeline_Dataflow

Activities

- Pipeline
 - Pipeline
 - Change Data Capture (preview)
 - Template gallery
 - Import from pipeline template
- Dataset
- Data flow
- Power Query
- Copy Data tool

Databricks

Data Lake Analytics

General

HDInsight

Iteration & conditionals

Machine Learning

Properties

General Related

Name *

Ass_8_part2_Q3_DataFlow_Pipeline

Description

3. Create a pipeline to remove the duplicate record available in the CSV file and save it as JSON file.

Annotations

+ New

Ass_8_Part2_Q3_Data...

ADLS_Generic_CSV_DS

Json_Ass8_part2_Q3_...

Ass_8_part2_Q3_D...

Validate

Debug

Add trigger

Data flow debug

Ass_8_part2_Q3_DataFlow_Pipeline

Ass_8_part2_Q3_DataFlow_Pipeline

General

Settings

Parameters

User properties

Name *

Ass_8_part2_Q3_DataFlow_Pipeline

Learn more

Description

3. Create a pipeline to remove the duplicate record available in the CSV file and save it as JSON file.a

Activity state

Activated

Deactivated

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Preview experience ☐ Off

Ass_8_Part2_Q3_Data... ADLS_Generic_CSV_DS Json_Ass8_part2_Q3_... Ass_8_part2_Q3_D...

Validate Debug Add trigger Data flow debug

Data flow

Ass_8_part2_Q3_DataFlow Pipeline

General Settings Parameters¹ User properties

Data flow * Ass_8_Part2_Q3_DataFlow Open New

Run on (Azure IR) * AutoResolveIntegrationRuntime

Compute size * Small

> Advanced

Logging level * Verbose Basic None

> Sink properties

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Publish all 4 Preview experience ☐ Off

Ass_8_Tough_Q1_Dat... Ass_8_Part2_Q3_Data... ADLS_Generic_CSV_DS Json_Ass8_part2_Q3_... Ass_8_part2_Q3_D...

Activities Validate Debug Add trigger Data flow debug

data Trigger test runs of the current pipeline without publishing your changes to the service

Move and transform

Copy data

Data flow

Azure Data Explorer

Azure Data Explorer C...

Databricks

Notebook

Jar

Python

Data Lake Analytics

U-SQL

General Settings Parameters¹ User properties

Data flow * Ass_8_Part2_Q3_DataFlow Open New

Run on (Azure IR) * AutoResolveIntegrationRuntime

Compute size * Small

> Advanced

Logging level * Verbose Basic None

> Sink properties

> Staging¹

Ass_8_Part2_Q3_Data... ADLS_Generic_CSV_DS Json_Ass8_part2_Q3... Ass_8_part2_Q3_D...

✓ Validate ⌚ Cancel options ⚡ Add trigger 🔵 Data flow debug ✓

Data flow

Ass_8_part2_Q3_DataFlow Pipeline

Parameters Variables Settings **Output**

Pipeline run ID: 232b33ca-4c94-4cf5-9a7c-324a489fe69c **Pipeline status:** In progress

All status Monitor in Azure Metrics Export to CSV

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration
Ass_8_part2_Q3_DataFlow_Pi...	Queued	Data flow	3/3/2024, 10:38:33 PM	Less than 1:

Search resources, services, and docs (G+)

Containers > landing >

Location: landing / Ass_8_output_files

Search blobs by prefix (cas...)

Show deleted objects

Name

- Ass_8_Q4_Product_Result
- Ass_8_Q5_ProductSingleRes...
- _SUCCESS
- part-00000-9a7b9495-5877...
- part-00000-a90c8241-fe0e-...
- part-00000-c9a42b9a-dbf7-...
- part-00053-9a7b9495-5877...
- part-00157-9a7b9495-5877...
- part-00160-9a7b9495-5877...
- part-merged.json

Ass_8_output_files/part-merged.json

Blob

Save Discard Download Refresh Delete

Overview Versions **Edit** Generate SAS

```
1 {"id": "3", "name": "Suresh_new", "city": "UK"}
2 {"id": "2", "name": "Ramesh_new", "city": "US"}
3 {"id": "4", "name": "ac", "city": "NY"}
4
```

Json Preview

Preview experience

Off

Ass_8_Part2_Q3_Data...

ADLS_Generic_CSV_DS

Json_Ass8_part2_Q3_...

Ass_8_part2_Q3_D...

Validate

Debug

Add trigger

Data flow debug

Data flow

Ass_8_part2_Q3_DataFlow Pipeline

Parameters

Variables

Settings

Output

Pipeline run ID: 0899f85c-5db3-4640-b3c7-e1cf7d0f2e0c

Pipeline status: Succeeded

View debug run consumption

All status

Monitor in Azure Metrics

Export to CSV

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration
Ass_8_part2_Q3_DataFlow_Pi...	Succeeded	Data flow	3/3/2024, 10:42:27 PM	2m 3s

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Publish all

Preview experience

Off

Ass_8_Tough_Q1_Dat...

Ass_8_Part2_Q3_Data...

ADLS_Generic_CSV_DS

Json_Ass8_part2_Q3_...

Ass_8_part2_Q3_D...

Validate

Debug

Add trigger

Data flow debug

Activities

Search activities

Move and transform

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

Data Lake Analytics

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Data flow

Ass_8_part2_Q3_DataFlow Pipeline

Parameters

Variables

Settings

Output

Pipeline run ID: 0899f85c-5db3-4640-b3c7-e1cf7d0f2e0c

Pipeline status: Succeeded

View debug run consumption

All status

Monitor in Azure Metrics

Export to CSV

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration
Ass_8_part2_Q3_DataFlow_Pi...	Succeeded	Data flow	3/3/2024, 10:42:27 PM	2m 3s

Preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

[Publish all](#) 4 Preview experience ☐ Off  

Ass_8_Tough_Q1_Dat... Ass_8_Part2_Q3_Data... ADLS_Generic_CSV_DS Json_Ass8_part2_Q3_... Ass_8_part2_Q3_D...

Activities  Search activities

- > Move and transform
- > Synapse
- > Azure Data Explorer
- > Azure Function
- > Batch Service
- > Databricks
- > Data Lake Analytics
- > General
- > HDInsight
- > Iteration & conditionals
- > Machine Learning
- > Power Query

29 J...

Validate Debug Add trigger Data flow debug 

Data flow  Ass_8_part2_Q3_DataFlow Pipeline 

Parameters Variables Settings **Output**

Pipeline run ID: 0899f85c-5db3-4640-b3c7-e1cf7d0f2e0c  

Pipeline status  Succeeded [View debug run consumption](#)

All status  [Monitor in Azure Metrics](#)  [Export to CSV](#) 

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration
Ass_8_part2_Q3_DataFlow_Pi...	 Succeeded	Data flow	3/3/2024, 10:42:27 PM	2m 3s

announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

ry    [Publish all](#) Preview experience ☐ Off  

sources  rces by name 

29

!_JOIN_Ingestion_Customer_J... !_Foreach_Example_2 !_Foreach_Example_2 !_Cust_YYYY_MM_DD !_Get_MetaData_Example !_Table_HighWaterMarkFile !ues_3 !ues_5_Foreach_Example2 !art2_Q1_Tough_pipeline !2_Pipeline_Dataflow !3_Pipeline_Dataflow !4_Pipeline_Dataflow !5_Pipeline_Dataflow

Activities  Search activities

- > Move and transform
- > Synapse
- > Azure Data Explorer
- > Azure Function
- > Batch Service
- > Databricks
- > Data Lake Analytics
- > General
- > HDInsight
- > Iteration & conditionals
- > Machine Learning
- > Power Query

Validate Debug Add trigger Data flow debug 

Data flow  Ass_8_part2_Q3_DataFlow Pipeline 

Parameters Variables Settings **Output**

Pipeline run ID: 0899f85c-5db3-4640-b3c7-e1cf7d0f2e0c  

Pipeline status  Succeeded [View debug run consumption](#)

All status  [Monitor in Azure Metrics](#)  [Export to CSV](#) 

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration
Ass_8_part2_Q3_DataFlow_Pi...	 Succeeded	Data flow	3/3/2024, 10:42:27 PM	2m 3s